

Answers to exercise 01

PART A

1	D	2	A	3	D	4	C	5	C
6	A	7	A	8	C	9	A	10	C
11	C	12	A	13	B	14	A	15	C
16	D	17	C	18	C	19	B	20	B
21	A	22	B						

PART B

1. (a) shared, (b) logically related (c) description
2. (a) internal schema, (b) conceptual schema, (c) external schema (以上三小题答案与顺序无关), (d) external schema, (e) internal schema (f) conceptual schema (以上两小题答案与顺序无关)
3. (a) DBA, (b) Database system analyst, (c) database designers, (d) application programmers, (e) end users (本题答案与顺序无关)
4. (a) external schemas, (b) conceptual schema, (c) conceptual schema, (d) external schema, (e) application programs (以上两小题答案与顺序无关), (f) external/conceptual.
5. (a) conceptual schema, (b) internal schema, (c) internal schema, (d) conceptual and external schemas, (e) conceptual/internal.
6. (a) data, (b) relationships between data, (c) constraints on the data (以上三小题答案与顺序无关), (d) structural part/data structure, (e) manipulative part/data manipulation, (f) integrity rules/data integrity constraint (以上三小题答案与顺序无关).

PART C.

- (a) Data: data is the physical symbols that represent different kinds of information.
- (b) Database: database is a shared collection of logically related data (and a description of this data), designed to meet the information needs of an organization.
- (c) Database management system: DBMS is a data management software (System Software) between users and operation system, it provide an environment that is both convenient and efficient for people to use in storing and retrieving data.
- (d) Database system: a system to organize, store, retrieve, manage, maintain data.
- (e) Data independence: the immunity of application programs to changes of the data structure.

PART D.

1.

Data independence refers to the immunity of application programs to changes of the data structure. There are two types of data independence in a database environment: logical data independence and physical data independence.

Logical data independence refers to the immunity of external schemas to changes in

conceptual schema. Physical data independence refers to the immunity of conceptual schemas to changes in internal schema.

Data independence has important significance, since the changes in the lower level of the database system do not incur the change in the higher level of the database system. This makes the database environment easier to be managed and used than file-based systems.

2.

A data model refers to an integrated collection of concepts for describing data, relationships between data, and constraints on the data.

A data model includes three components: the data structure, the data operations, and the integrity rules.